

STATISTICAL INFERENCE

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HW 3: Part I

- **Important Note:** This homework has been intentionally designed to be more manageable and straightforward in consideration of your upcoming midterm exam. While it includes some computer assignments, they are simple and easy to solve.
- If you have any questions about the homework, don't hesitate to use the class group to ask questions or drop an email to the HW Authors.
- This course aims to equip you with the skills to tackle all problems in this domain and encourages you to engage in independent research. Utilize your learnings to extend beyond the classroom teachings where necessary.

Problem 1: Method of Moments in Quality Control

A manufacturing company produces electronic components whose lifetimes (in thousands of hours) follow a Gamma distribution with shape parameter $\alpha > 0$ and scale parameter $\beta > 0$. The probability density function is:

$$f(t; \alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} t^{\alpha-1} e^{-t/\beta}, \quad t > 0$$

The quality control department needs to estimate both parameters to assess product reliability and predict warranty costs. They randomly select n components from production and test them until failure, recording lifetimes $T_1, T_2, \dots, T_n$.

Part (a): Derive the method-of-moments (MoM) estimators $\tilde{\alpha}$ and $\tilde{\beta}$ for the shape and scale parameters. Express your estimators in terms of the sample mean $\bar{T}$ and sample variance S^2 .

Part (b): Prove that the MoM estimator $\tilde{\beta}$ is a consistent estimator of β . That is, show that $\tilde{\beta} \xrightarrow{P} \beta$ as $n \rightarrow \infty$.

Part (c): The engineering specifications indicate that the shape parameter is $\alpha = 3$ for this component type. Using this information, derive a simplified MoM estimator for β based only on the sample mean. Determine whether this simplified estimator is unbiased for β .

Part (d): Suppose the company tests $n = 50$ components and obtains $\bar{T} = 7.2$ thousand hours and $S^2 = 10.8$ (thousand hours)². Calculate the MoM estimates of α and β . Based on these estimates, what is the expected lifetime of a component?

Problem 2: Maximum Likelihood Estimation in Clinical Trials

A pharmaceutical company is conducting a clinical trial for a new drug designed to reduce recovery time from a specific illness. Historical data suggests that recovery times (in days) follow an exponential distribution with rate parameter $\lambda > 0$, where the probability density function is:

$$f(t; \lambda) = \lambda e^{-\lambda t}, \quad t > 0$$

The company recruits n patients and records their recovery times $T_1, T_2, \dots, T_n$. The goal is to estimate the rate parameter λ to determine the drug's effectiveness and expected recovery time $\mu = 1/\lambda$.

Part (a): Write down the likelihood function $L(\lambda)$ and the log-likelihood function $\ell(\lambda)$ for the observed data. What is the interpretation of the likelihood function in this context?

Part (b): Derive the maximum likelihood estimator (MLE) $\hat{\lambda}_{MLE}$ by finding the critical point of the log-likelihood function. Show all steps including the first-order condition.

Part (c): Compute the score function $U(\lambda)$ and verify that $E[U(\lambda)] = 0$ when evaluated at the true parameter value. What is the interpretation of the score function?

Part (d): Calculate the Fisher information $I(\lambda)$ using two methods:

- (i) Direct calculation: $I(\lambda) = -E \left[\frac{\partial^2 \ell(\lambda)}{\partial \lambda^2} \right]$
- (ii) Via the variance of the score: $I(\lambda) = \text{Var}(U(\lambda))$

Verify that both methods yield the same result and interpret what Fisher information tells us about parameter estimation.

Part (e): Show that the MLE $\hat{\lambda}_{MLE}$ is:

- (i) Unbiased for λ
- (ii) Consistent for λ
- (iii) Asymptotically efficient (achieves the Cramér-Rao lower bound)

Part (f): Derive the method-of-moments (MoM) estimator $\tilde{\lambda}_{MoM}$ for this problem. Compare it with the MLE and discuss whether they are the same or different. What does this tell us about the relationship between MoM and MLE?

Part (g): In a trial with $n = 100$ patients, the observed recovery times yield $\sum_{i=1}^{100} T_i = 450$ days.

- (i) Calculate the MLE estimate $\hat{\lambda}$
- (ii) Estimate the expected recovery time
- (iii) Using the asymptotic normality of the MLE, construct an approximate 95% confidence interval for λ

Problem 3: Comparing Estimators: MoM vs MLE Simulation Study

(Use R or Python) Building on the theoretical analysis of Method of Moments and Maximum Likelihood Estimation, this problem investigates their finite-sample behavior through simulation.

Consider the Gamma distribution with shape parameter $\alpha = 3$ and scale parameter $\beta = 2$ (as in the quality control problem). The probability density function is:

$$f(t; \alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} t^{\alpha-1} e^{-t/\beta}, \quad t > 0$$

Tasks:

(a) **Simulation Setup:**

- Generate $N = 1000$ datasets, each containing n observations from $\text{Gamma}(\alpha = 3, \beta = 2)$ for different sample sizes: $n \in \{10, 25, 50, 100, 200\}$
- For each dataset, compute:
 - Method of Moments estimators: $\tilde{\alpha}_{MoM}$ and $\tilde{\beta}_{MoM}$ using

$$\tilde{\alpha} = \frac{\bar{T}^2}{S^2}, \quad \tilde{\beta} = \frac{S^2}{\bar{T}}$$
 - Maximum Likelihood estimators: $\hat{\alpha}_{MLE}$ and $\hat{\beta}_{MLE}$ (use numerical optimization)

(b) **Performance Metrics:**

Calculate and compare for both estimators of β across all sample sizes:

- **Bias:** $\text{Bias}(\hat{\beta}) = \frac{1}{N} \sum_{i=1}^N \hat{\beta}_i - \beta$
- **Variance:** $\text{Var}(\hat{\beta}) = \frac{1}{N-1} \sum_{i=1}^N (\hat{\beta}_i - \bar{\hat{\beta}})^2$
- **Mean Squared Error:** $\text{MSE}(\hat{\beta}) = \text{Bias}^2(\hat{\beta}) + \text{Var}(\hat{\beta})$

Present results in a comprehensive table.

(c) **Visualizations:**

Create the following plots:

- Sampling distributions:** For $n \in \{25, 100\}$, create overlaid histograms of $\tilde{\beta}_{\text{MoM}}$ and $\hat{\beta}_{\text{MLE}}$ with vertical line at true $\beta = 2$
- Bias comparison:** Line plot of bias vs. sample size for both estimators
- MSE comparison:** Plot MSE vs. sample size on log-log scale for both estimators
- Box plots:** Side-by-side box plots showing distribution of estimates for $n = 25$ and $n = 100$

Deliverables:

- Well-commented code
- Summary table of bias, variance, MSE
- All required visualizations with proper labels and legends
- Detailed written interpretations of results

Problem 4: Inference for Air Quality Monitoring

Air quality researchers are analyzing hourly measurements of PM2.5 concentration (in $\mu\text{g}/\text{m}^3$) collected from a major metropolitan area. Assume the measurements follow an i.i.d. sample from a normal distribution:

$$X_1, X_2, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$$

where μ represents the true average PM2.5 concentration and σ^2 is the true population variance.

A random sample of $n = 36$ hourly observations produced the following summary statistics:

$$\bar{X} = 28.4, \quad S^2 = 49.$$

Answer the following questions.

- Point Estimation:** Define the point estimators for μ and σ^2 . Briefly explain in the context of this study the difference between *point estimation* and *interval estimation*.
- Confidence Interval Derived from MLE:** Using the fact that the MLE of μ for a Normal distribution is $\bar{X}$, construct a 95% confidence interval for μ assuming the variance σ^2 is *known* and equal to S^2 .
- Exact Confidence Interval (Variance):** Construct a 95% exact confidence interval for σ^2 using the chi-square method.
- Approximate Confidence Interval (Mean):** Now assume that σ^2 is *unknown*. Construct a 95% approximate confidence interval for μ using the *t*-distribution.
- Bootstrap Method (Practical Implementation):**

In this part, you will construct a 95% bootstrap confidence interval for the mean μ using the bootstrap percentile method.

- (i) Briefly describe the bootstrap concept and why it may be preferred when theoretical distributional assumptions may not hold.
- (ii) Write a pseudo-algorithm summarizing the bootstrap procedure to estimate a confidence interval for μ using the observed sample.
- (iii) Compare the bootstrap interval conceptually to the analytical intervals obtained in parts (b) and (d).

Problem 5: Coding Confidence Intervals

(Use R or Python) The following are the systolic blood pressure measurements for two groups of men and women randomly selected for a heart study:

Men:

128.35, 160.34, 133.74, 138.12, 91.00, 97.43, 128.58, 148.78, 150.65, 110.96, 135.7, 118.77, 147.1, 107.2, 122.46, 129.36, 158.14, 102.72, 136.59, 146.02, 105.88, 111.24, 131.22, 124.6, 137.85, 136.46, 145.31, 166.71, 158.66, 108.63, 103.11, 149.29

Women:

116.62, 137.15, 106.07, 172.58, 151.33, 98.73, 136.11, 149.9, 140.8, 98.58, 158.4, 97.97, 117.99, 126.53, 128.67, 126.57, 124.3, 120.39, 150.08, 143.05, 130.18, 108.04, 136.39, 124.94, 136.86, 143.03, 128.58, 142.51, 151.68, 120.94

1. Visualize the Data:

- Create histograms to view the distribution of systolic blood pressures for both groups.
- Generate boxplots to compare the central tendencies and variability of the two groups.

2. Statistical Analysis:

- Calculate basic statistics such as mean, variance, and range for each group.
- Interpret these statistics to discuss potential differences in blood pressure variability and central tendency between the groups.

3. Confidence Interval Construction:

- (a) Construct a 95% confidence interval for the difference in mean systolic blood pressures between men and women. Interpret the result.
- (b) Is there any difference in mean systolic blood pressures between men and women? Explain the conclusion.

4. Interpretation:

- Interpret the results of the histograms, boxplots, and confidence interval.
- Discuss the implications of these findings in the context of the heart study.

Problem 6: True Coverage of Different CI (Optional)

(Use R or Python) This problem examines the true coverage probability for three different methods of making confidence intervals for a sample proportion.

Given: Sample size $n = 60$ and $p = 0.4$.

For each of the following methods, find which outcomes x result in confidence intervals that capture p and compute the coverage probability.

1. Normal from maximum likelihood estimate, $\hat{p} = \frac{x}{n}$, $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$, with the interval $\hat{p} \pm 1.96SE$.
2. Normal from adjusted maximum likelihood estimate, $\hat{p} = \frac{x+2}{n+4}$, $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{(n+4)}}$, with the interval $\hat{p} \pm 1.96SE$.
3. Within $z^2/2$ of the maximum likelihood loglikelihood. (For this method, use the provided function `log1.ci.p()` which returns the lower and upper bounds of a 95% confidence interval given n and x . You can graph the loglikelihood using `glog1.p()` for n , x , and $z = 1.96$ to see if the returned values make sense.)

Problem 7: Playing around with the hypothesis

Suppose that a single observation X is taken from a uniform density on $[0, \theta]$, and consider testing $H_0 : \theta = 1$ versus $H_1 : \theta = 2$.

1. Find a test that has significance level $\alpha = 0$. What is its power?
2. For $0 < \alpha < 1$, consider the test that rejects when $X \in [0, \alpha]$. What is its significance level and power?
3. What is the significance level and power of the test that is rejected when $X \in [1 - \alpha, 1]$?
4. Find another test that has the same significance level and power as the previous one.
5. Does the likelihood ratio test determine a unique rejection region?
6. What happens if the null and alternative hypotheses are interchanged- $H_0 : \theta = 2$ versus $H_1 : \theta = 1$?

Problem 8: Fundamentals of Hypothesis Testing: Quality Control (Optional)

A manufacturing company produces precision ball bearings with a target diameter of $\mu_0 = 10.0$ mm. The production process has a known standard deviation of $\sigma = 0.5$ mm. Quality control engineers randomly sample $n = 25$ bearings from a production batch and measure their diameters $X_1, X_2, \dots, X_{25}$, which are assumed to be i.i.d. $N(\mu, \sigma^2)$ where $\sigma = 0.5$ is known but μ is unknown.

Part (a): The engineers want to test whether the production process is meeting specifications.

- (i) Formulate appropriate null and alternative hypotheses if they want to detect any deviation from the target (either too large or too small). Classify these hypotheses as simple or composite, and identify this as a one-sided or two-sided test.
- (ii) Now suppose they are specifically concerned about bearings being undersized (which causes safety issues). Reformulate the hypotheses for this scenario and classify the test type.

Part (b): For the two-sided test in Part (a)(i):

- (i) Define Type I and Type II errors in this context.
- (ii) What are the practical consequences of each error type for the company and customers?
- (iii) If the company sets $\alpha = 0.05$, what does this mean in practical terms?

Part (c): Consider the test statistic $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$ for the two-sided test.

- (i) What is the null distribution of Z ? Derive this result.
- (ii) For significance level $\alpha = 0.05$, determine the critical region (rejection region) for H_0 .
- (iii) What is the acceptance region for H_0 ?
- (iv) Interpret what these regions mean in terms of the sample mean $\bar{X}$.

Problem 9: P-values and Confidence Intervals in Hypothesis Testing

A manufacturing company produces precision ball bearings with a target diameter of $\mu_0 = 10.0$ mm. The production process has a known standard deviation of $\sigma = 0.5$ mm. Quality control engineers randomly sample $n = 25$ bearings and obtain $\bar{x} = 9.75$ mm. Assume $X_1, \dots, X_{25} \sim N(\mu, \sigma^2)$ with $\sigma = 0.5$ known.

They test $H_0 : \mu = 10.0$ vs. $H_1 : \mu \neq 10.0$ at $\alpha = 0.05$.

Part (a): Using the test statistic $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$:

- (i) Calculate the test statistic z .
- (ii) Make a decision: should H_0 be rejected at $\alpha = 0.05$?
- (iii) Calculate the p-value for this test. Define what the p-value represents and interpret its meaning in this context.
- (iv) How does the p-value relate to the decision in part (ii)?

Part (b): Construct a 95% confidence interval for the true mean diameter μ .

- (i) Calculate the confidence interval.
- (ii) Explain the duality between confidence intervals and hypothesis testing: How can you use this confidence interval to test $H_0 : \mu = 10.0$ versus $H_1 : \mu \neq 10.0$ at $\alpha = 0.05$?
- (iii) Verify that your confidence interval approach gives the same conclusion as the hypothesis test in Part (a).
- (iv) What is the relationship between the confidence level $(1 - \alpha)$ and the significance level α ?

Problem 10: Power Analysis in Hypothesis Testing (Optional)

A manufacturing company produces precision ball bearings with a target diameter of $\mu_0 = 10.0$ mm. The production process has a known standard deviation of $\sigma = 0.5$ mm. They test $H_0 : \mu = 10.0$ vs. $H_1 : \mu \neq 10.0$ using a sample of $n = 25$ bearings at significance level $\alpha = 0.05$. Assume $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with $\sigma = 0.5$ known.

Part (a): Power analysis for the two-sided test.

- (i) Define the power of a test. What does it measure?
- (ii) Calculate the power of this test when the true mean is $\mu_1 = 9.6$ mm.
- (iii) Calculate the Type II error probability β for $\mu_1 = 9.6$ mm.
- (iv) Interpret what this power means for quality control purposes.

Part (b): Effect of sample size on power.

- (i) How would the power change if the sample size increased to $n = 100$? Calculate the new power for $\mu_1 = 9.6$ mm.
- (ii) Compare the power at $n = 25$ and $n = 100$. What does this tell you about the importance of sample size?

Part (c): Understanding the relationship between α , β , and n .

- (i) If the company reduces α from 0.05 to 0.01 (keeping $n = 25$), what happens to the power? Explain conceptually and verify numerically for $\mu_1 = 9.6$.
- (ii) Describe the shape and key features of the power curve $\beta(\mu)$ as a function of μ for $\mu \in [9.0, 11.0]$ with $n = 25$ and $\alpha = 0.05$.
- (iii) What are the implications of the α -power trade-off for practical decision-making?

Problem 11: Power Analysis and Hypothesis Testing Performance

(Use **R** or **Python**) Test $H_0 : \mu = 10.0$ vs. $H_1 : \mu \neq 10.0$ with $\sigma = 0.5$, $\alpha = 0.05$.

Part (a): Power Curves and Sample Size

- For $n \in \{10, 25, 50, 100, 200\}$, compute power over $\mu \in [9.0, 11.0]$:

$$\text{Power}(\mu) = \Phi\left(\frac{\mu - 10}{\sigma/\sqrt{n}} - 1.96\right) + \Phi\left(-\frac{\mu - 10}{\sigma/\sqrt{n}} - 1.96\right)$$

Plot all curves. Mark 0.80 power crossings.

- Create heat map: required n for power $\{0.70, 0.80, 0.90\}$ and effect size $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$

Part (b): Error Regions Visualization

For $n = 25$, $\mu_1 = 9.6$:

- Plot $N(10.0, 0.01)$ and $N(9.6, 0.01)$
- Shade: Type I error (α), Type II error (β), Power ($1 - \beta$)
- Compute and label all probabilities

Part (c): Trade-offs and Verification

For $n = 25$, $\mu_1 = 9.6$:

- Plot power vs. α for $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$
- Run $N = 5,000$ simulations:
 - Verify empirical α under H_0 ($\mu = 10$)
 - Verify empirical power under H_1 ($\mu = 9.6$)

Part (d): Interpretation

- Explain why decreasing α reduces power. Show this in your visualization.
- Why is the power curve symmetric around $\mu_0 = 10$?
- Do simulations match theory? If not, explain discrepancies.
- For quality control: recommend α choice considering cost of false alarms vs. missed defects.

Deliverables:

- Python or R implementations
- Theory vs. simulation table

Problem 12: Optimality Theory: LRT, Neyman-Pearson, and UMP Tests

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables from a probability distribution with parameter θ .

Part (a): Likelihood Ratio Tests

Consider $X_1, \dots, X_n \sim \text{Exponential}(\theta)$ with pdf $f(x; \theta) = \theta e^{-\theta x}$, $x > 0$. Test $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$ where $\theta_1 > \theta_0$.

- Derive and simplify the likelihood ratio $\frac{L(\theta_1)}{L(\theta_0)}$.
- Show the rejection region has form $T(\mathbf{x}) < c$ and identify $T(\mathbf{x})$.

Part (b): Neyman-Pearson Lemma

- (i) State the Neyman-Pearson Lemma.
- (ii) For the exponential case in Part (a), derive the most powerful test of size α .
- (iii) Find the constant c such that the test has size exactly α .
- (iv) Calculate the power function $\beta(\theta_1)$.

Part (c): Uniformly Most Powerful Tests

Consider $H_0 : \theta = \theta_0$ vs. $H_1 : \theta > \theta_0$ for the exponential distribution.

- (i) Define a Uniformly Most Powerful (UMP) test.
- (ii) Show that the test from Part (a) is UMP for $H_1 : \theta > \theta_0$.
- (iii) Does a UMP test exist for $H_1 : \theta \neq \theta_0$? Why or why not?

Part (d): Comparison of Testing Procedures

For $X_1, \dots, X_n \sim N(\theta, 1)$, test $H_0 : \theta \leq \theta_0$ vs. $H_1 : \theta > \theta_0$.

- (i) Derive the likelihood ratio test and its rejection region.
- (ii) Show this is the UMP test.
- (iii) Verify that both approaches yield the same rejection region.

Part (e): Asymmetry in Hypothesis Testing

- (i) Why is H_0 treated asymmetrically from H_1 in the Neyman-Pearson framework?
- (ii) Why do we say "fail to reject H_0 " rather than "accept H_0 "?

Problem 13: GLRT (Optional)

Let $X_1, \dots, X_n$ be i.i.d. from an exponential distribution with unknown mean θ . Consider testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

1. Derive the generalized likelihood ratio test for this setup.
2. Find the asymptotic distribution of $-2 \log \Lambda$ under H_0 , where Λ is the likelihood ratio.

Problem 14: Learn about the t-dist before too late (Optional)

(Use R or Python) This problem examines a t-distribution with 20 degrees of freedom.

1. Plot a graph of a t-distribution with 20 degrees of freedom and a standard normal curve from -4 to 4.
2. Find the area to the right of 2 under each curve.
3. Find the 0.90 quantile of each curve.
4. Find the 0.95 quantile of each curve.
5. Find the 0.975 quantile of each curve.